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Gear device

Abstract

A gear device for a gear system for an aircraft includes: an input stage epicyclic gear comprising an input stage sun, an input stage carrier and an input stage ring; and an output stage epicyclic gear comprising an output stage sun, an output stage carrier and an output stage ring. A first input stage component of the input stage sun, the input stage carrier and the input stage ring is fixed to a corresponding first output stage component of the output stage sun, the output stage carrier and the output stage ring, wherein a second input stage component of the input stage sun, the input stage carrier and the input stage ring is fixed to a corresponding second output stage component of the output stage sun, the output stage carrier and the output stage ring. The second input stage component is different from the first input stage component.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to European Patent Application No. 22306724.0 filed Nov. 23,

2022, the entire contents of which is incorporated herein by reference.

(2) This disclosure relates to a gear device and a gear system both of which can be used on an aircraft.

BACKGROUND

(3) Gear devices are used in gear systems in aircraft to convert low torque high speed motion (e.g. from an electric motor) to high torque low speed motion (e.g. to a flight control component). For such gear devices it is desirable to achieve as high a gear ratio as possible for a compact gear device. It is also desirable to reduce design complexity and improve manufacturability of gear devices.

SUMMARY

(4) According to a first aspect, there is provided a gear device for a gear system for an aircraft, the gear device comprising: an input stage epicyclic gear comprising an input stage sun, an input stage carrier and an input stage ring; and an output stage epicyclic gear comprising an output stage sun, an output stage carrier and an output stage ring, wherein a first input stage component of the input stage sun, the input stage carrier and the input stage ring is fixed to a corresponding first output stage component of the output stage sun, the output stage carrier and the output stage ring, wherein a second input stage component of the input stage sun, the input stage carrier and the input stage ring is fixed to a corresponding second output stage component of the output stage sun, the output stage carrier and the output stage ring, wherein the second input stage component is different from the first input stage component, wherein a third input stage component of the input stage sun, the input stage carrier and the input stage ring is held stationary, wherein the third input stage component is different from the first input stage component and the second input stage component, wherein a third output stage component is connectable to an output shaft, the third output stage component corresponding to the third input stage component.

(5) By connecting two epicyclic gears together in this way and fixing an input stage component which corresponds to the output stage component, a high gear ratio may be achieved in a compact gear device with a simple design which is easy to manufacture.

(6) The gear device may comprise the output shaft.

(7) The first input stage component may be the input stage sun and the first output stage component may be the output stage sun.

(8) The second input stage component may be the input stage ring and the second output stage component may be the output stage ring.

(9) The third input stage component may be the input stage carrier and the third output stage component may be the output stage carrier. This may provide a convenient, on-axis output from the gear device.

(10) The first input stage component may be connectable to an input shaft. The gear device may comprise the input shaft.

(11) The input stage epicyclic gear may have an input stage fixed carrier train ratio, R_i , and the output stage epicyclic gear may have an output stage fixed carrier train ratio, R_o , wherein $R_i \neq R_o$ and $|R_i - R_o|/R_o < 1$. $|R_i - R_o|/R_o$ may be less than 0.5. $|R_i - R_o|/R_o$ may be approximately 0.2. $|R_i - R_o|/R_o$ may be approximately 0.19. The input stage sun may comprise an input stage sun number of teeth, N_{is} , and the output stage sun may comprise an output stage sun number of teeth, N_{os} , wherein $N_{is} \neq N_{os}$ and $|N_{is} - N_{os}|/N_{os} < 1$. $|N_{is} - N_{os}|/N_{os}$ may be less than 0.5. $|N_{is} - N_{os}|/N_{os}$ may be approximately 0.2. N_{is} may be approximately 30 and N_{os} may be approximately 25.

(12) Having kinematic similarity but not identity between the input stage epicyclic gear and the output stage epicyclic gear, whilst fixing an input stage component which corresponds to the output stage component, may permit a higher gear ratio to be achieved (e.g. with the fixed input stage component having a speed of zero and the corresponding output stage having a non-zero but close to zero speed).

(13) According to a second aspect, there is provided a gear device for a gear system for an aircraft,

the gear device comprising: an input stage epicyclic gear comprising an input stage sun, an input stage carrier and an input stage ring; and an output stage epicyclic gear comprising an output stage sun, an output stage carrier and an output stage ring, wherein the input stage sun is connected to the output stage sun, wherein the input stage ring is connected to the output stage ring, wherein the input stage carrier is held stationary, wherein the output stage carrier is connectable to an output shaft.

(14) The gear device of the second aspect may comprise any of the features described above with respect to the gear device of the first aspect.

(15) According to a third aspect, there is provided a gear system, the gear system comprising the gear device described above (with respect to the first aspect and/or the second aspect) and a system input shaft connectable to an actuator and a system output shaft connectable to flight control component, the input stage epicyclic gear configured to receive torque from the system input shaft and the output stage epicyclic gear configured to transmit torque to the output shaft.

(16) According to a fourth aspect, there is provided a flight control system comprising the gear system described above, the actuator connected to the system input shaft and the flight control component connected to the system output shaft.

(17) The actuator may be an electric motor.

(18) The flight control component may comprise an aileron. The flight control component may comprise a flap. The flight control component may comprise a rudder. The flight control component may comprise a spoiler, The flight control component may comprise an elevator. The flight control component may comprise a slat.

(19) According to a fifth aspect, there is provided an aircraft comprising a flight control system as described above.

(20) The aircraft of the fifth aspect may comprise any of the features of the flight control system of the fourth aspect. The flight control system of the fourth aspect may comprise any of the features of the gear system of the third aspect. The gear system of the third aspect may comprise any of the features of the gear device of the first aspect and/or the second aspect.

Description

BRIEF DESCRIPTION OF FIGURES

(1) Embodiments will be further described and explained by way of example with reference to the accompanying drawings in which:

(2) FIG. 1 shows a schematic of an aircraft comprising a flight control system; and

(3) FIG. 2 shows a schematic of a gear device.

DETAILED DESCRIPTION

(4) With reference to FIG. 1, there is shown an aircraft 2. The aircraft comprises a flight control system 4. The flight control system 4 comprises an actuator 6. The actuator 6 is an electric motor 6. In particular, the actuator 6 is a high speed low torque electric motor 6.

(5) The flight control system 4 comprises a gear system 8. The flight control system 4 comprises a flight control component 10. The flight control component 10 comprises a wing flap. In other examples, the flight control component 10 comprises other components, including an aileron and/or a rudder.

(6) The gear system 8 comprises a system input shaft 12. The system input shaft 12 is connected to the actuator 6. The system input shaft 12 is configured to receive a torque from the actuator 6 and rotate in response to the torque.

(7) The gear system 8 comprises a system output shaft 14. The system output shaft 14 is connected to the flight control component 10. The system output shaft 14 is configured to apply a torque to a shaft (not shown) of the flight control component 10 to control the aircraft 2.

- (8) While only a single flight control system is depicted in FIG. 1, it will be understood that in practice the aircraft may include several flight control systems (e.g. ailerons, spoilers, rudders, elevators, slats and/or flaps), with each flight control system comprising one or more flight control components and associated gear systems as described here.
- (9) The gear system **8** comprises first, second and third gear devices **100a-c**. As described below, each of the first, second and third gear devices **100a-c** provides a gearing stage and is configured to increase a torque and reduce a speed of a rotation.
- (10) The first, second and third gear devices **100a-c** are connected in series. The first gear device **100a** is connected to the system input shaft **12** to receive a first rotational motion from the system input shaft **12**. The first gear device **100a** converts the first rotational motion to a second rotational motion, the second rotational motion being of a higher torque and lower speed than the first rotational motion.
- (11) The second gear device **100b** is connected to the first gear device **100a** to receive the second rotational motion from the first gear device **100a**. The second gear device **100b** converts the second rotational motion to a third rotational motion, the third rotational motion being of higher torque and lower speed than the second rotational motion.
- (12) The third gear device **100c** is connected to the second gear device **100b** to receive the third rotational motion from the second gear device **100b**. The third gear device **100c** converts the third rotational motion to a fourth rotational motion, the fourth rotational motion being of higher torque and lower speed than the third rotational motion.
- (13) The third gear device **100c** is connected to the system output shaft **14**. The third gear device **100c** provides the fourth rotational motion to the system output shaft **14**, such that the system output shaft **14** provides the fourth rotational motion to the flight control component **10**.
- (14) In other examples, the gear system **8** comprises a different number of gear devices **100a-c** to provide a different number of stages (e.g. more than three devices and stages).
- (15) Referring to FIG. 2, there is shown a gear device **100**, which may be the first gear device **100a**, the second gear device **100b** and/or the third gear device **100c**. The gear device **100** comprises an input stage epicyclic gear **102**.
- (16) The input stage epicyclic gear **102** comprises an input stage sun **104**. The input stage sun **104** is a gear. The input stage sun **104** comprises external teeth (i.e. extending radially outwardly from the input stage sun **104**).
- (17) The input stage epicyclic gear **102** comprises an input stage carrier **106**. The input stage carrier **106** is connected to input stage planets **108**. The input stage planets **108** are gears. The input stage planets **108** comprise external teeth (i.e. extending radially outwards from a central axis of each input stage planet **108**). The input stage planets **108** mesh with the input stage sun **104**. In particular, the teeth of the input stage planets **108** mesh with the teeth of the input stage sun **104**.
- (18) In the present example the input stage epicycle gear **102** comprises three input stage planets **108**. The input stage planets **108** are evenly circumferentially displaced about the input stage sun **104**. Each the input stage planet **108** is individually rotatable (e.g. about its central axis) relative to the input stage carrier **106**.
- (19) The input stage epicyclic gear **102** comprises an input stage ring **110**. The input stage ring **110** is a gear. The input stage ring **110** comprises internal teeth (i.e. extending radially inwardly from the input stage ring **110**). The input stage ring **110** meshes with the input stage planets **108**. In particular, the teeth of the input stage ring **110** mesh with the teeth of the input stage planets **108**.
- (20) The gear device **100** comprises an output stage epicyclic gear **112**.
- (21) The output stage epicyclic gear **112** comprises an output stage sun **114**. The output stage sun **114** is a gear. The output stage sun **114** comprises external teeth (i.e. extending radially outwardly from the output stage sun **114**).
- (22) The output stage epicyclic gear **112** comprises an output stage carrier **116**. The output stage carrier **116** is connected to output stage planets **118**. The output stage planets **118** are gears. The

output stage planets **118** comprise external teeth (i.e. extending radially outwards from a central axis of each output stage planet **118**). The output stage planets **118** mesh with the output stage sun **114**. In particular, the teeth of the output stage planets **118** mesh with the teeth of the output stage sun **114**.

(23) In the present example the output stage epicycle gear **112** comprises three output stage planets **118**. The output stage planets **118** are evenly circumferentially displaced about the output stage sun **114**. Each output stage planet **118** is individually rotatable (e.g. about its central axis) relative to the output stage carrier **116**. The output stage planets **118** are able to orbit the output stage sun **112**, causing rotation of the output stage carrier **116**.

(24) The output stage epicyclic gear **112** comprises an output stage ring **120**. The output stage ring **120** is a gear. The output stage ring **120** comprises internal teeth (i.e. extending radially inwardly from the output stage ring **120**). The output stage ring **120** meshes with the output stage planets **118**. In particular, the teeth of the output stage ring **120** mesh with the teeth of the output stage planets **118**.

(25) The gear device **100** comprises an input shaft **122** and an output shaft **124**. Depending on the position of the gear device **100** within the gear system **8** (i.e. which stage it is), the input shaft **122** is connected to an output shaft of another gear device or to the actuator **6** to receive rotational motion therefrom. Depending on the position of the gear device **100** with the gear system **8** (i.e. which stage it is), the output shaft **124** is connected to an input shaft of another gear device or to the flight control component **10** to transmit rotation motion therefrom.

(26) Each component of the input stage epicyclic gear **102** corresponds to a component of the output stage epicyclic gear **112**. The input stage sun **104** corresponds to the output stage sun **114**. The input stage carrier **106** corresponds to the output stage carrier **116**. The input stage ring **110** corresponds to the output stage ring **120**.

(27) The input stage epicyclic gear **102** and the output stage epicyclic gear **112** are arranged concentrically with each other. In particular, the input stage sun **104** is concentric with the output stage sun **114**. The input stage carrier **106** is concentric with the output stage carrier **116**. The input stage ring **110** is concentric with the output stage ring **120**.

(28) A first input stage component of the input stage sun **104**, the input stage carrier **106** and the input stage ring **110** is fixed to a corresponding first output stage component of the output stage sun **114**, the output stage carrier **116** and the output stage ring **120**. The first input stage component is connected to the input shaft **122**. In the present example, the first input stage component is the input stage sun **104**. The first output stage component is the output stage sun **114**.

(29) As such, the input stage sun **104** is rigidly connected to the input shaft **122** (in the present example, formed as an integral component), such that the input stage sun **104** is configured to rotate with the input shaft **122**. The output stage sun **114** is rigidly connected to the input stage sun **104**. The output stage sun **114** is rigidly connected to the input shaft **122** (via the input stage sun **104**). The output stage sun **114** is therefore configured to rotate with the input stage sun **104** and the input shaft **122**.

(30) A second input stage component of the input stage sun **104**, the input stage carrier **106** and the input stage ring **110** is fixed to a corresponding second output stage component of the output stage sun **114**, the output stage carrier **116** and the output stage ring **120**. The second input stage component is a different component from the first input stage component. In the present example, the second input stage component is the input stage ring **110**. The second output stage component is the output stage ring **120**.

(31) As such, the input stage ring **110** is rigidly connected to the output stage ring **120** (in the present example, formed as an integral component), such that the output stage ring **120** is configured to rotate with the input stage ring **110**.

(32) A third input stage component of the input stage sun **104**, the input stage carrier **106** and the input stage ring **110** is held stationary. The third input stage component is a different component

from the first input stage component and the second input stage component. A third output stage component of output stage sun **114**, the output stage carrier **116** and the output stage ring **120** is connected to the output shaft **124**. The third output stage component corresponds to the third input stage component. In the present example, the third input stage component is the input stage carrier **106** and the third output stage component is the output stage carrier **116**.

(33) As such, rotation of the input stage carrier **106** is prevented, also preventing the orbiting of the input stage planets **118** about the input stage sun **104**. The output stage carrier **116** is rigidly connected to the output shaft **124**, such that the output shaft **124** is configured to rotate with the output stage carrier **116** (i.e. on orbiting of the output stage planets **118** about the output stage sun **114**).

(34) The input stage sun **104** comprises an input stage sun number of teeth, N_{is} . In the present example, $N_{is}=30$. The input stage ring **110** comprises an input stage ring number of teeth, N_{ir} . In the present example, $N_{ir}=66$. It will be understood that the number of teeth of each input stage planet **108** is defined by N_{is} and N_{ir} , so discussion of the number of teeth of each input stage planet **108** is omitted.

(35) The input stage epicyclic gear **102** is defined by a fixed carrier train ratio, R_i , where $R_i=-N_{ir}/N_{is}$. In the present example, $R_i=-2.2$ (2 significant figures).

(36) The output stage sun **114** comprises an output stage sun number of teeth, N_{os} . In the present example, $N_{os}=25$. The output stage ring **120** comprises an output stage ring number of teeth, N_{or} . In the present example, $N_{or}=68$. It will be understood that the number of teeth of each output stage planet **108** is defined by N_{os} and N_{or} .

(37) The output stage epicyclic gear **112** is defined by a fixed carrier train ratio, R_o , wherein $R_o=-N_{or}/N_{os}$. In the present example, $R_o=2.7$ (2 significant figures).

(38) In use, the input shaft **122** rotates, causing rotation of the input stage sun **104**. The rotation of the input stage sun **104** causes rotation of the input stage planets **108** about their own axes. Rotation of the input stage carrier **106** is prevented, so the input stage planets **108** cannot orbit the input stage sun **104**.

(39) The rotation of the input stage planets **108** causes rotation of the input stage ring **110**.

(40) The output stage sun **114** rotates at the same speed as the input stage sun **104**. The output stage ring **120** rotates at the same speed as the input stage ring **110**.

(41) The fixed carrier train ratios of the input and output epicyclic gears **102**, **112** are different (due to different kinematics of the input and output epicyclic gears resulting from different numbers of teeth), which means that the rotation of the output stage ring **120** and the output stage sun **114** causes rotation of the output stage carrier **116**. This in turn causes rotation of the output shaft **124**.

(42) Because the input stage carrier is fixed **106**, the output stage carrier **116** is only able to rotate slowly. This results in output rotation at the output shaft **124** which is significantly lower speed and higher torque than at the input shaft **122**, resulting in a high gear ratio for a compact gear device **100**.

(43) In the present example this effect is amplified by the input stage epicyclic gear and the output stage epicyclic gear being kinematically similar but not identical, which, due to the input stage carrier having zero rotation, results in the speed of the rotation of the output stage carrier **116** being close to zero but not zero.

(44) The applicant has found that gear ratio is further increased where the input stage fixed carrier train ratio, R_i , is close but not the same as the output stage fixed carrier train ratio, R_o , in particular where $|R_i-R_o|/R_o < 1$. In the present example, $|R_i-R_o|/R_o=0.19$ (2 sf).

(45) Additionally, the applicant has found that the gear ratio is further increased where the input stage sun number of teeth, N_{is} , is similar but not identical to the output stage sun number of teeth, N_{os} , in particular, where $|N_{is}-N_{os}|/N_{os} < 1$. In the present example, $|N_{is}-N_{os}|/N_{os}=0.20$ (2 sf).

(46) In the present example, the gear ratio is 15.

(47) In general, by increasing the kinematic similarity between the input stage epicyclic gear and

the output stage epicyclic gear, the gear ratio increases. However, increasing the kinematic similarity may also require tighter manufacturing tolerances. Increasing the kinematic similarity, particularly where tight manufacturing tolerances cannot be achieved, may increase contact friction between the gears, reducing efficiency of the system. Design of the gear device may also be constrained by gears needing to have an integer number of teeth.

(48) Various aspects of the gear device disclosed in the various embodiments may be used alone, in combination, or in a variety of arrangements not specifically discussed in the embodiments described in the foregoing and this disclosure is therefore not limited in its application to the details and arrangement of components set forth in the foregoing description or illustrated in the drawings. For example, aspects described in one embodiment may be combined in any manner with aspects described in other embodiments. Although particular embodiments have been shown and described, it will be obvious to those skilled in the art that changes and modifications may be made without departing from the invention in its broader aspects. The scope of the following claims should not be limited by the embodiments set forth in the examples, but should be given the broadest reasonable interpretation consistent with the description as a whole.

Claims

1. A gear arrangement for an aircraft, comprising a plurality of gear systems, each of the gear system comprising: an input shaft; an output shaft; an input stage epicyclic gear comprising an input stage sun that is configured to rotate and connected to the input shaft, an input stage carrier that is stationary, an input stage ring that is configured to rotate, and input stage planetary gears supported by the input stage carrier and disposed between the input stage sun and input stage ring; and an output stage epicyclic gear comprising an output stage sun that is rotationally fixed to the input stage sun, an output stage carrier that is configured to rotate and connected directly or indirectly to the output shaft, an output stage ring that is rotationally fixed to the input stage ring, and output stage planetary gears supported by the output stage carrier and disposed between the output stage sun and output stage ring; wherein the input stage epicyclic gear and the output stage epicyclic gear are arranged concentrically with each other such that the input and output stage suns are concentric with each other, the input and output stage carriers are concentric with each other, and the input and output stage rings are concentric with each other; and a first fixed carrier train ratio of the input stage epicyclic gear and a second fixed carrier train ratio of the output stage epicyclic gear are configured such that when the input shaft rotates at a first speed, the input and output stage suns rotate together at the first speed, the input and output stage rings rotate together at a second speed, and the output stage carrier rotates at a third speed that is less than the first speed, whereby the output shaft rotates slower than the input shaft, and when the output shaft is connected directly to the output stage carrier, the output shaft rotates at the third speed, wherein the plurality of gear systems are connected to one another in series, wherein the input shaft is connected to the input stage sun of a first one of the gear systems and the output shaft is connected to the output stage carrier of a last one of the gear systems, wherein each of the gear systems is configured to reduce rotational speed and increase torque relative to a previous one of the gear systems so that, in operation, the output shaft rotates at a reduced speed relative to the input shaft and delivers greater torque than the input shaft.

2. The gear arrangement of claim 1, wherein the first fixed carrier train ratio is R_i , the second fixed carrier train ratio is R_o , and wherein $R_i \neq R_o$ and $|R_i - R_o|/R_o < 1$.

3. The gear arrangement of claim 2, wherein $|R_i - R_o|/R_o < 0.5$.

4. The of claim 2, wherein $|R_i - R_o|/R_o \approx 0.19$.

5. The gear arrangement of claim 1, wherein the input stage sun comprises an input stage sun number of teeth, N_{is} , and the output stage sun comprises an output stage sun number of teeth, N_{os} , wherein $N_{is} \neq N_{os}$ and $|N_{is} - N_{os}|/N_{os} < 1$.

6. The gear arrangement of claim 5, wherein $|N_{is}-N_{os}|/N_{os}<0.5$.
 7. The gear arrangement of claim 5, wherein $N_{is}\approx 30$ and $N_{os}\approx 25$.
 8. The of claim 5, wherein $|N_{is}-N_{os}|/N_{os}\approx 0.2$.
 9. A flight control system comprising: the gear arrangement of claim 1; an actuator coupled to the input shaft; and a flight control component connected to the output shaft.
 10. The flight control of claim 9, wherein the actuator is an electric motor.
 11. The flight control of claim 9, wherein the flight control component comprises one or more of an aileron, a flap, a spoiler, an elevator, a slat, or a rudder.
 12. An aircraft comprising: the flight control of claim 11.
 13. An aircraft comprising: one or more of an aileron, a flap, a spoiler, an elevator, a slat or a rudder; the flight control of claim 1, wherein the flight control component comprises the one or more of the aileron, the flap, the spoiler, the elevator, the slat or the rudder.
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